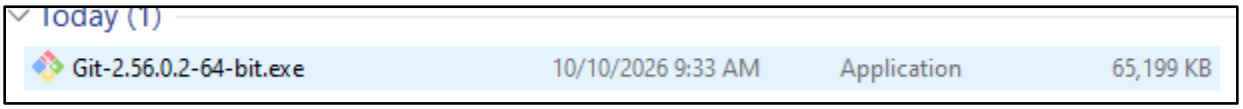
Fatima Jinnah Women University



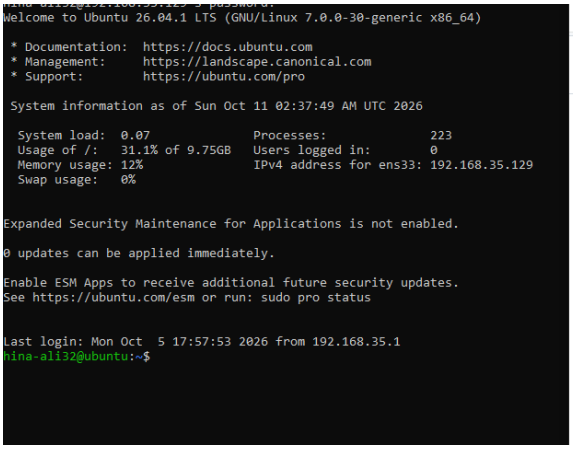
Department of Computer Science
DevOps Lab # 2

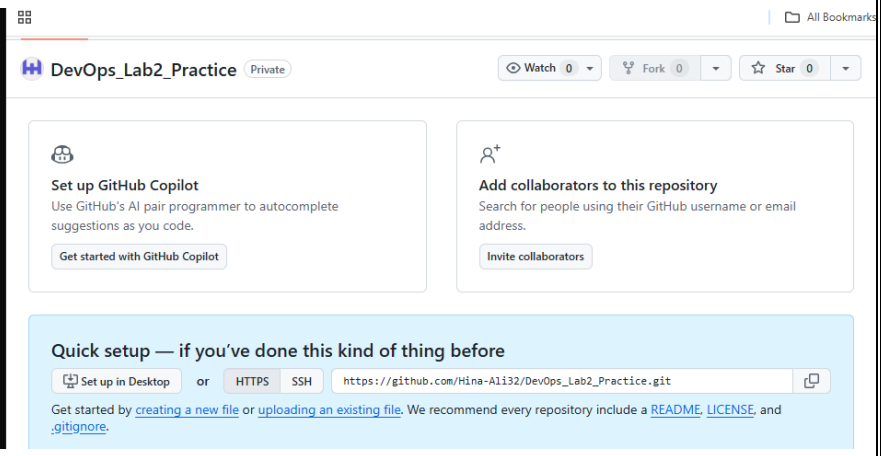
Name: Hina Ali
Registration no: 2023-BCS-040
Course code: CS-649
Section: BCS VII(B)
GitHub Username: Hina-Ali32

Getting Started



Task 1: Create Private GitHub Repository



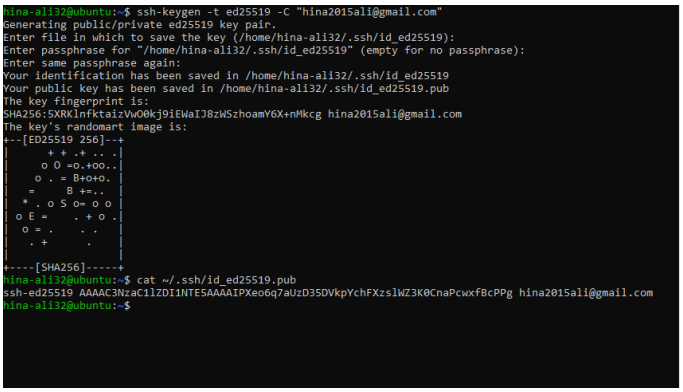


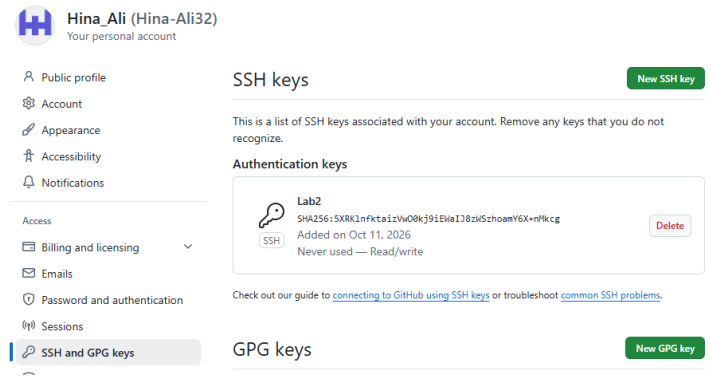
Shows the new DevOps_Lab2_Practice repository marked Private on GitHub, next to the terminal prompt.

Task 2: Connect Repository via SSH

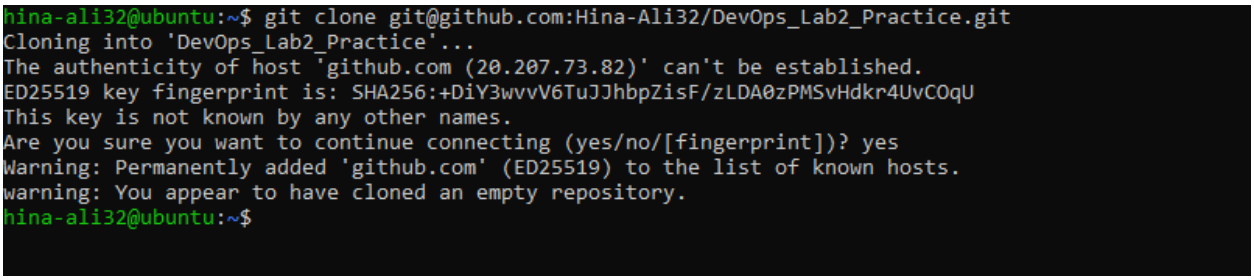
```
hina-ali32@ubuntu:~$ ssh-keygen -t ed25519 -C "hina2015ali@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/hina-ali32/.ssh/id_ed25519):
Enter passphrase for "/home/hina-ali32/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/hina-ali32/.ssh/id_ed25519
Your public key has been saved in /home/hina-ali32/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:5XRKlnfktaizVwO0kj9iEWaIJ8zWSzhoamY6X+nMkcg hina2015ali@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|      + + .+ .. .|
|     o O =o.+oo..|
|    o . = B+o+o. |
|   =     B +=..  |
|  * . o S o= o o |
| o E =    . + o .|
|  o = .    . .   |
|   . +      .    |
|                 |
+----[SHA256]-----+
hina-ali32@ubuntu:~$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIPXeo6q7aUzD35DVkpYchFXzslWZ3K0CnaPcwxfBcPPg hina2015ali@gmail.com
hina-ali32@ubuntu:~$
```

Shows an ed25519 SSH key pair generated on the Ubuntu server with ssh-keygen, followed by cat ~/.ssh/id_ed25519.pub displaying only the public key. The private key is not displayed.



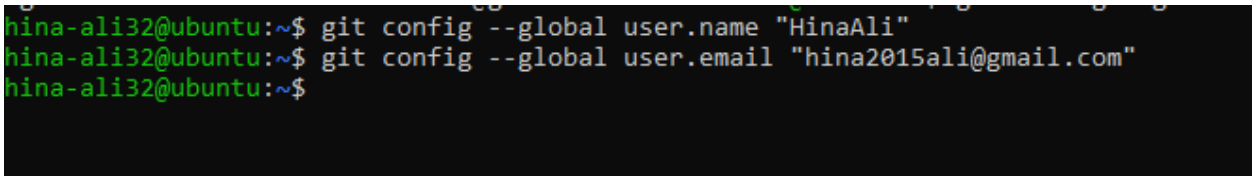


Shows the public key added to GitHub under **SSH and GPG keys** with the title Lab02 Ubuntu VM, next to the terminal that generated it.

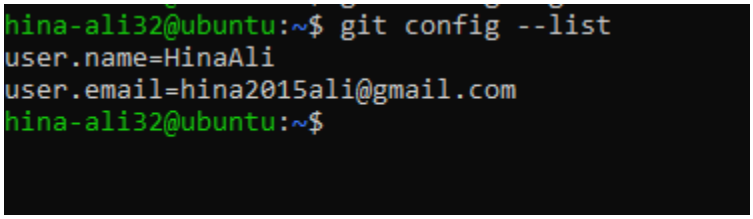


Shows git clone git@github.com:<username>/DevOps_Lab2_Practice.git completing over SSH. GitHub is added to the known hosts, and Git warns that the cloned repository is empty.

Task 3: Configure Git Username and Email



Shows git config --global user.name and git config --global user.email setting the identity used for all commits.



Shows git config --list with the configured user.name and user.email in the output, verifying that the identity was saved.

Task 4: Explore the .git Folder

```
hina-ali32@ubuntu:~$ cd DevOps_Lab2_Practice
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ ls -a .git
.  ..  HEAD  config  description  hooks  info  objects  refs
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows ls -a .git listing the hidden contents of the repository (HEAD, config, description, hooks, info, objects, refs), which hold the history, logs and configuration of the repository.

Task 5: Local Repository Management

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ rm -rf .git
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git status ==> error why?
fatal: not a git repository (or any of the parent directories): .git
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows rm -rf .git removing the repository data, followed by git status returning fatal: not a git repository. The error appears because Git stores all repository information in .git, so the folder is no longer a repository.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/hina-ali32/DevOps_Lab2_Practice/.git/
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git init creating a new empty Git repository in the project folder.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Initial commit"
[master (root-commit) 8be1c58] Initial commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows README.md created with echo, staged with git add, and committed with the message “Initial commit”.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git remote add origin git@github.com:Hina-Ali32/DevOps_Lab2_Practice.git
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git branch -M main
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 234 bytes | 117.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git remote add origin, git branch -M main and git push -u origin main connecting the local repository to GitHub and uploading the first commit.

Task 6: File Status & Staging

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ echo "Adding some test notes" >> notes.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        error
        notes.txt

nothing added to commit but untracked files present (use "git add" to track)
```

Shows git status listing notes.txt under “Untracked files”, meaning the file exists but is not tracked yet.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add notes.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Add notes.txt"
[main 796354b] Add notes.txt
 1 file changed, 1 insertion(+)
 create mode 100644 notes.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows notes.txt staged with git add and committed as “Add notes.txt”, confirmed by 1 file changed, 1 insertion(+) and create mode 100644 notes.txt.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ echo "Second note" >> notes.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

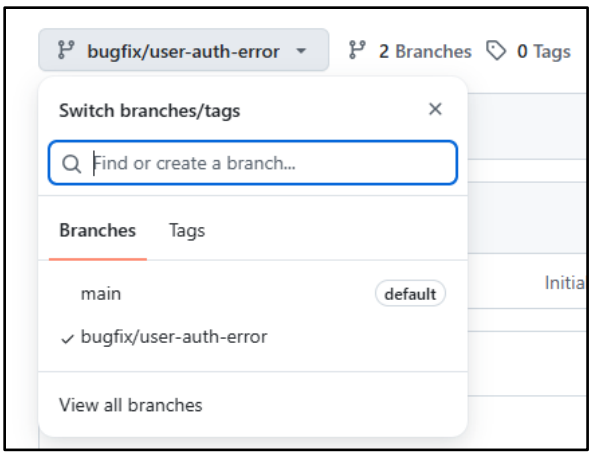
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   notes.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        error

no changes added to commit (use "git add" and/or "git commit -a")
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add notes.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Update notes.txt"
[main 3b56643] Update notes.txt
 1 file changed, 1 insertion(+)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git status listing notes.txt as “modified” under “Changes not staged for commit” after a second line was added. And Shows the modified notes.txt staged and committed as “Update notes.txt”.

Task 7: Branch Creation Using GitHub GUI



Shows the branch bugfix/user-auth-error created from main in the GitHub web interface, selected in the branch dropdown, with the repository listing two branches.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git pull origin bugfix/user-auth-error
From github.com:Hina-Ali32/DevOps_Lab2_Practice
 * branch            bugfix/user-auth-error -> FETCH_HEAD
 * [new branch]      bugfix/user-auth-error -> origin/bugfix/user-auth-error
Already up to date.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git branch -a
* main
  remotes/origin/bugfix/user-auth-error
  remotes/origin/main
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git pull origin bugfix/user-auth-error fetching the new branch to the server, followed by git branch -a listing remotes/origin/bugfix/user-auth-error

Task 8: Branch Creation and Push Using Git Bash

Shows git checkout -b feature/db-connection creating the branch and git push origin feature/db-connection publishing it, confirmed by [new branch] feature/db-connection –

>feature/db-connection

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git branch
* main
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Already on 'main'
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b feature/db-connection
Switched to a new branch 'feature/db-connection'
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin feature/db-connection
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 577 bytes | 82.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/db-connection' on GitHub by visiting:
remote:      https://github.com/Hina-Ali32/DevOps_Lab2_Practice/pull/new/feature/db-connection
remote:
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
 * [new branch]      feature/db-connection -> feature/db-connection
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Task 9: Branching & Merging

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b feature-1
Switched to a new branch 'feature-1'
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git checkout -b feature-1 creating the branch and switching to it.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ printf 'def greet(name):\n    return "Hello, " + name\n' >> main.py
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add main.py
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Add new function to main.py"
[feature-1 d5348b0] Add new function to main.py
 1 file changed, 2 insertions(+)
 create mode 100644 main.py
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows a function added to main.py and committed on feature-1 with the message “Add new function to main.py

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git merge feature-1
Updating 3b56643..d5348b0
Fast-forward
 main.py | 2 ++
 1 file changed, 2 insertions(+)
 create mode 100644 main.py
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

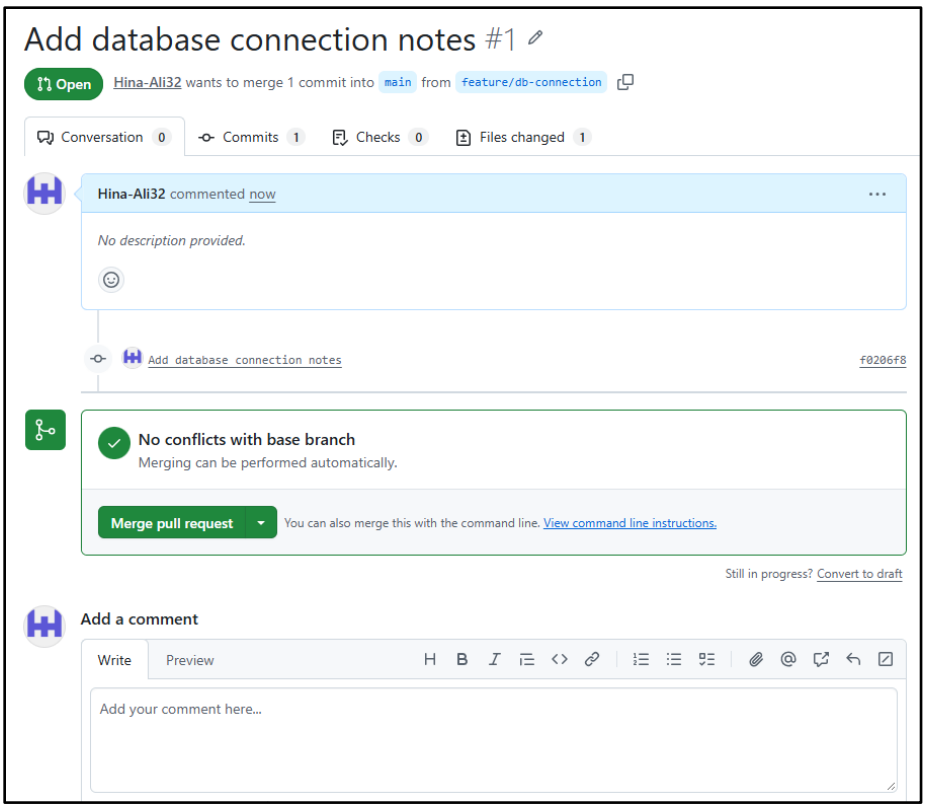
Shows git checkout main and git merge feature-1, bringing the main.py changes into main.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 360 bytes | 120.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   8be1c58..d5348b0  main -> main
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin feature-1
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-1' on GitHub by visiting:
remote:      https://github.com/Hina-Ali32/DevOps_Lab2_Practice/pull/new/feature-1
remote:
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
 * [new branch]      feature-1 -> feature-1
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

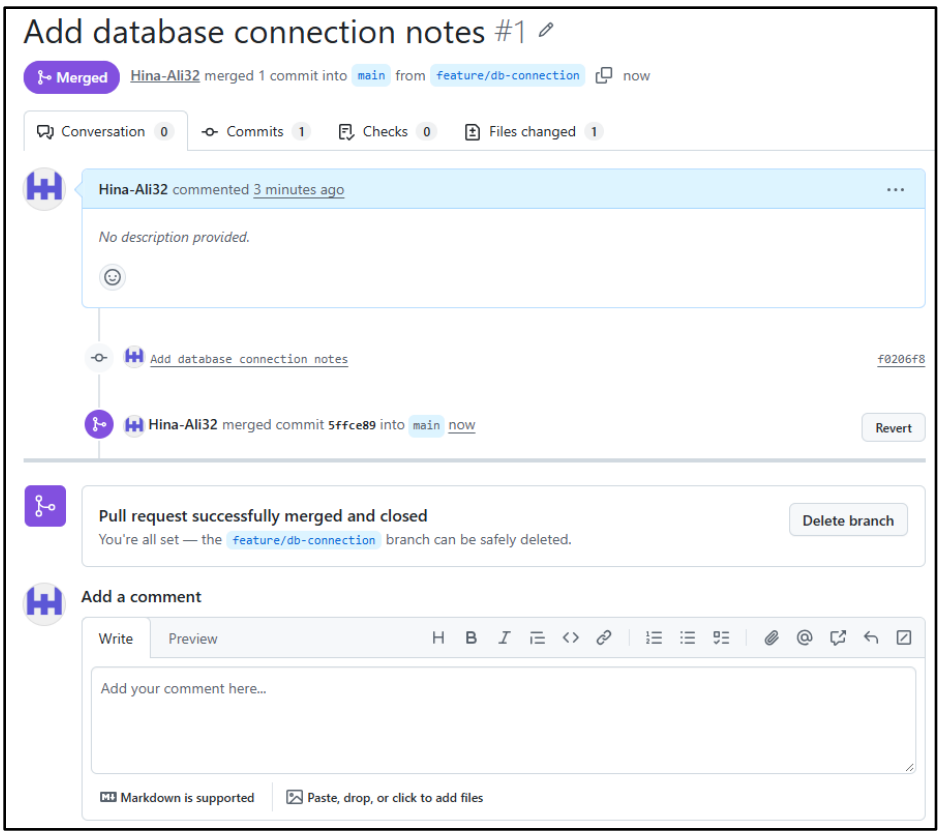
- Shows main and feature-1 both pushed to GitHub with git push origin main and git push origin feature-1.

Task 10: Pull Request and Branch Review (GitHub GUI)

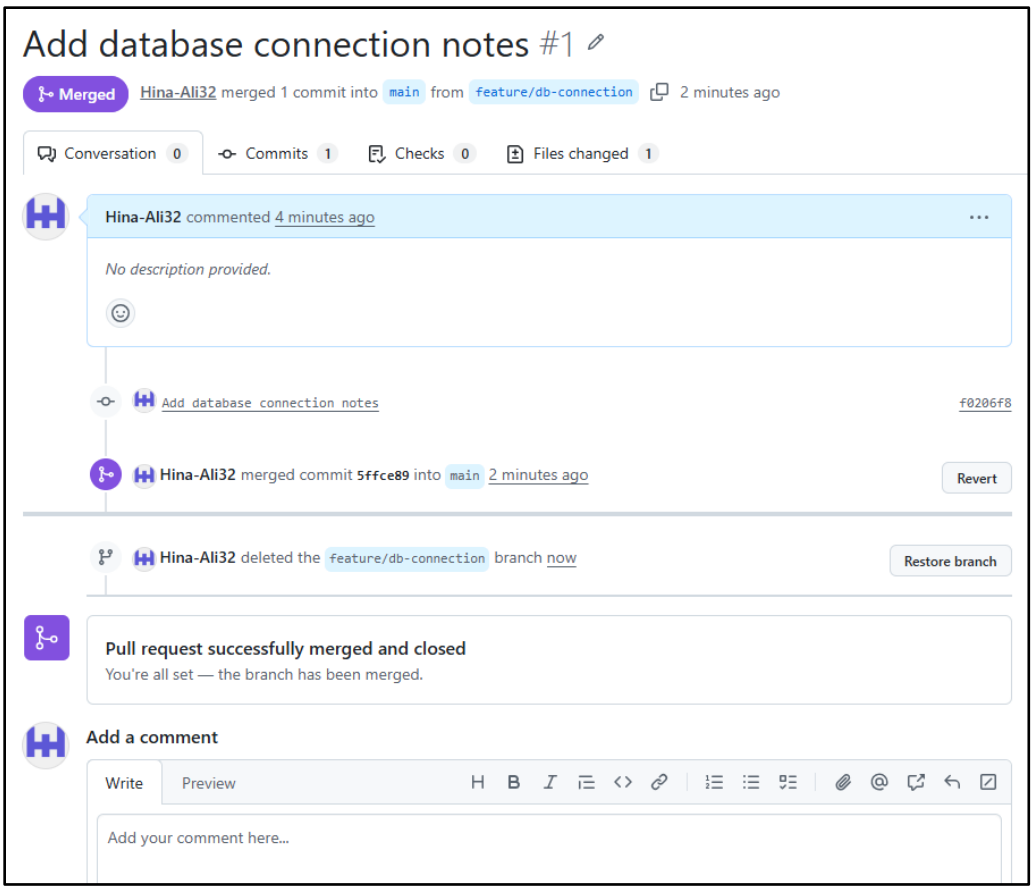
Shows pull request #1 “Add database connection notes” opened from feature/db-connection into main, with the status **Open** and “No conflicts with base branch”.



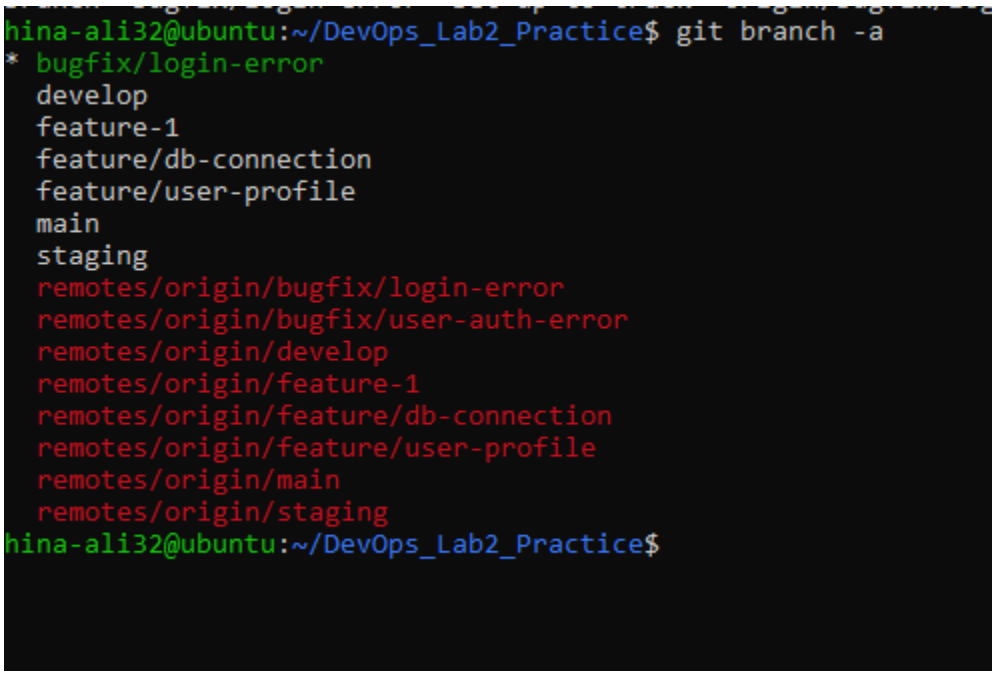
Shows the pull request reviewed and merged through the GitHub interface, with the purple **Merged** badge.



Shows the feature/db-connection branch deleted after the merge, with the Restore branch button visible.



Task 11: Detailed Branch Strategy (Develop/Staging)



Shows git branch -a listing develop, staging, feature/user-profile and bugfix/login-error alongside main.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout develop
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git merge --no-ff feature/user-profile -m "Merge feature/user-profile into develop"
Merge made by the 'ort' strategy.
 profile.md | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 profile.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git merge --no-ff bugfix/login-error -m "Merge bugfix/login-error into develop"
Merge made by the 'ort' strategy.
 login_fix.md | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 login_fix.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin develop
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 508 bytes | 169.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   5ffce89..bac02ec  develop -> develop
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout staging
Switched to branch 'staging'
Your branch is up to date with 'origin/staging'.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git merge --no-ff develop -m "Merge develop into staging"
Merge made by the 'ort' strategy.
 login_fix.md | 1 +
 profile.md   | 1 +
 2 files changed, 2 insertions(+)
 create mode 100644 login_fix.md
 create mode 100644 profile.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin staging
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 225 bytes | 225.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   5ffce89..5bd3f37  staging -> staging
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

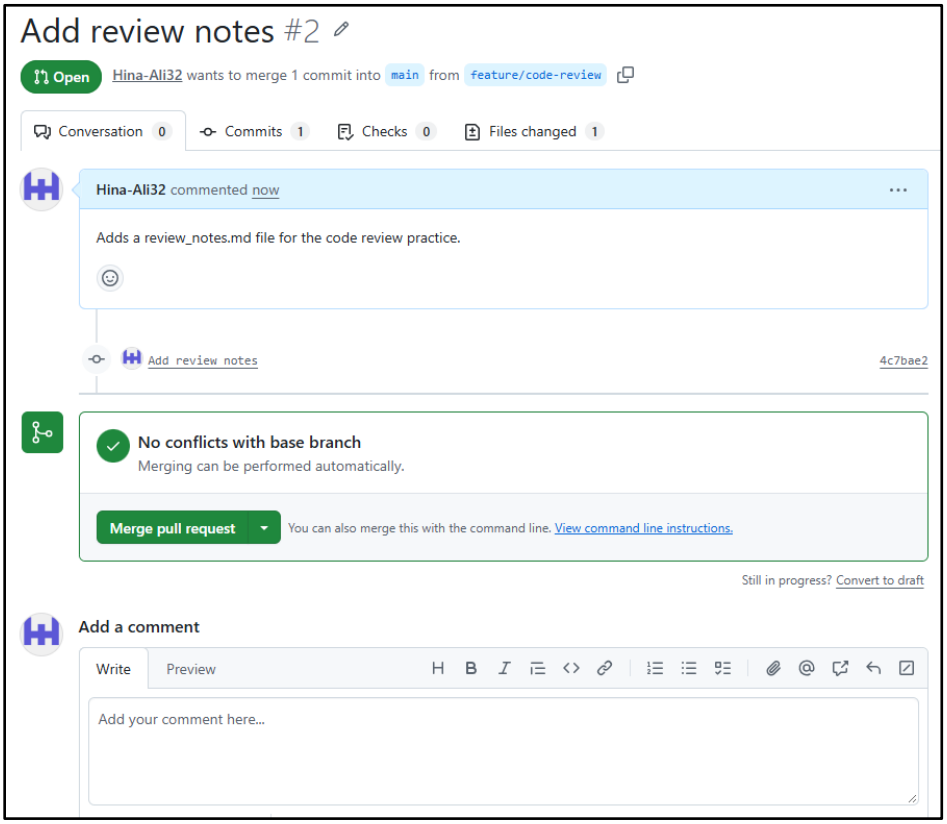
Shows feature/user-profile and bugfix/login-error merged into develop, then develop merged into staging, with each merge pushed to GitHub.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git merge --no-ff staging -m "Merge staging into main"
Merge made by the 'ort' strategy.
 login_fix.md | 1 +
 profile.md   | 1 +
 2 files changed, 2 insertions(+)
 create mode 100644 login_fix.md
 create mode 100644 profile.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 219 bytes | 109.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   5ffce89..1862267  main -> main
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

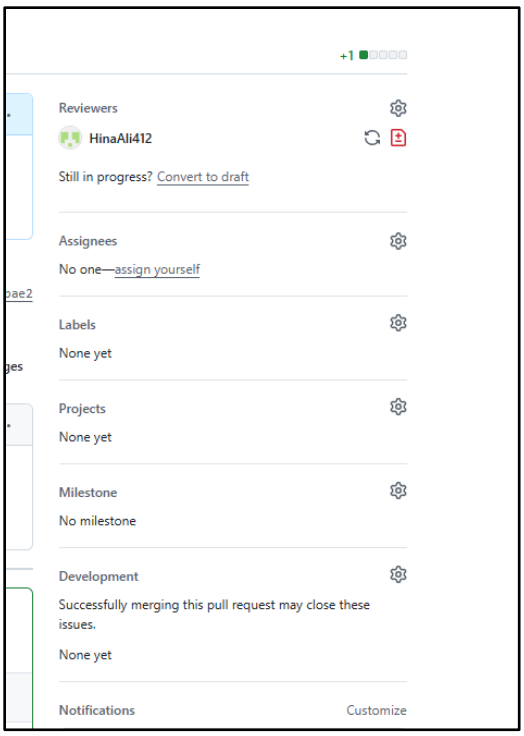
Shows staging merged into main with the message “Merge staging into main” and pushed to GitHub, completing the flow from feature branches to production.

Task 12: Code Review Workflow

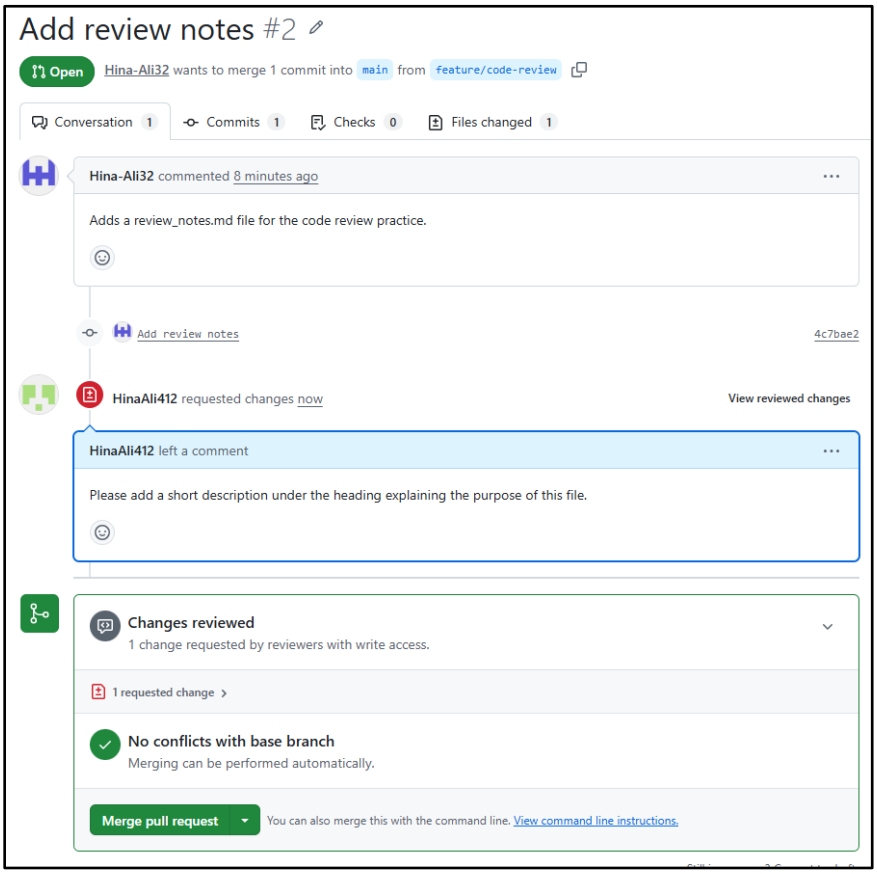
Shows pull request #2 “Add review notes” from feature/code-review into main, with a title and a description summarizing the change.



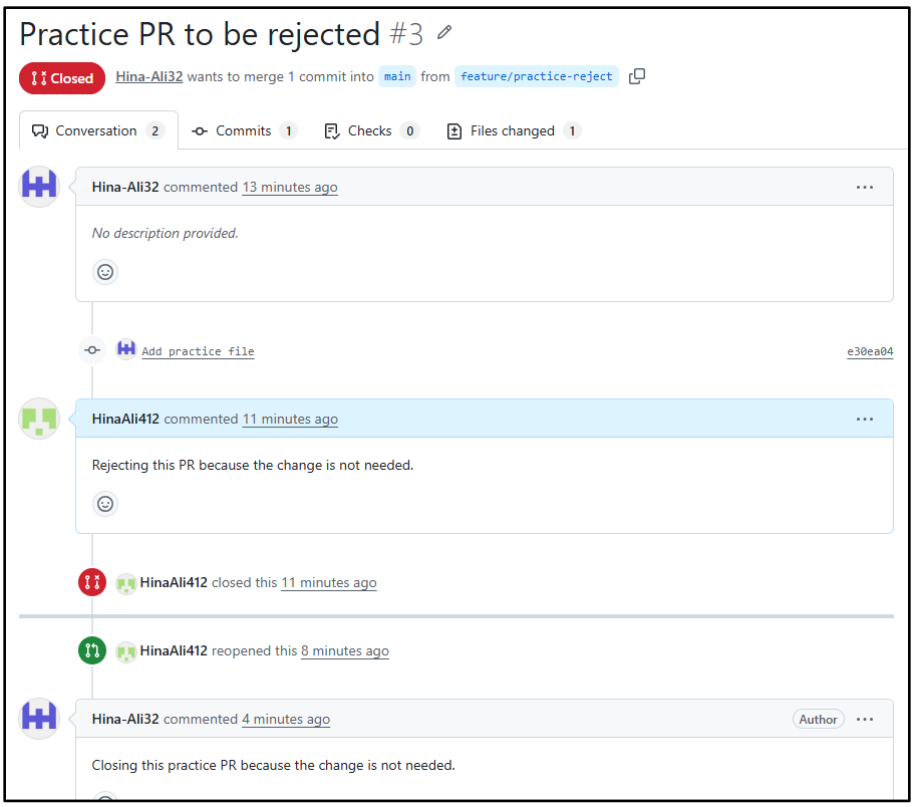
Shows the second account HinaAli412 listed under Reviewers in the sidebar of pull request #2.


+1
Reviewers
HinaAli412
Still in progress? Convert to draft
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone
Development
Successfully merging this pull request may close these issues.
None yet
Notifications
Customize

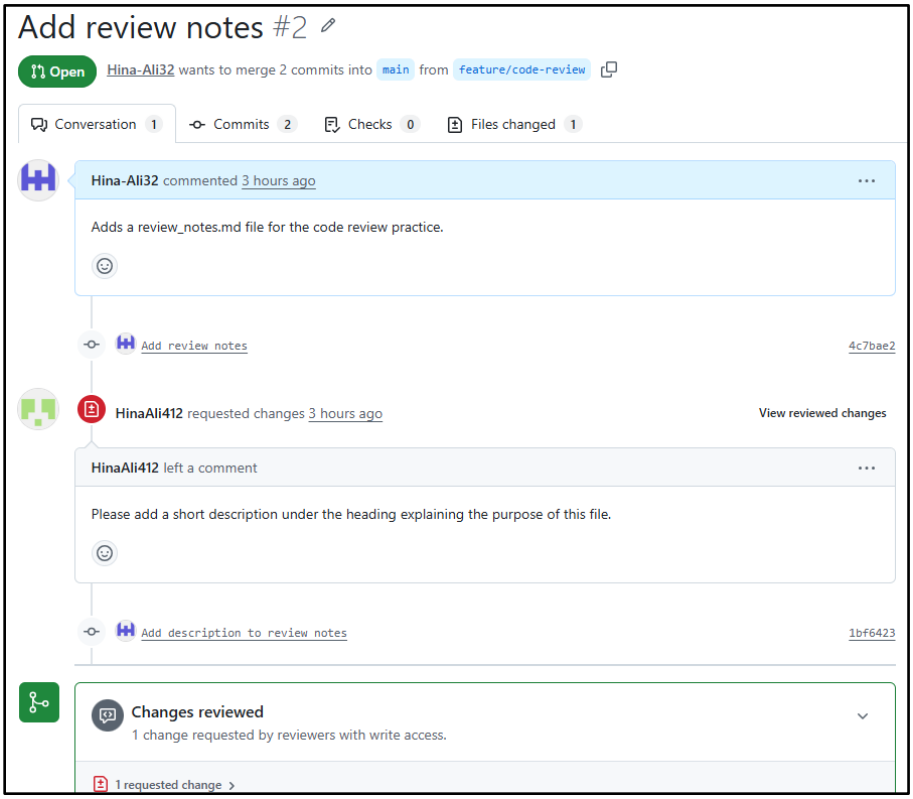
Shows the reviewer submitting **Request changes** with a comment asking for a short description under the heading, and the status “1 change requested”.



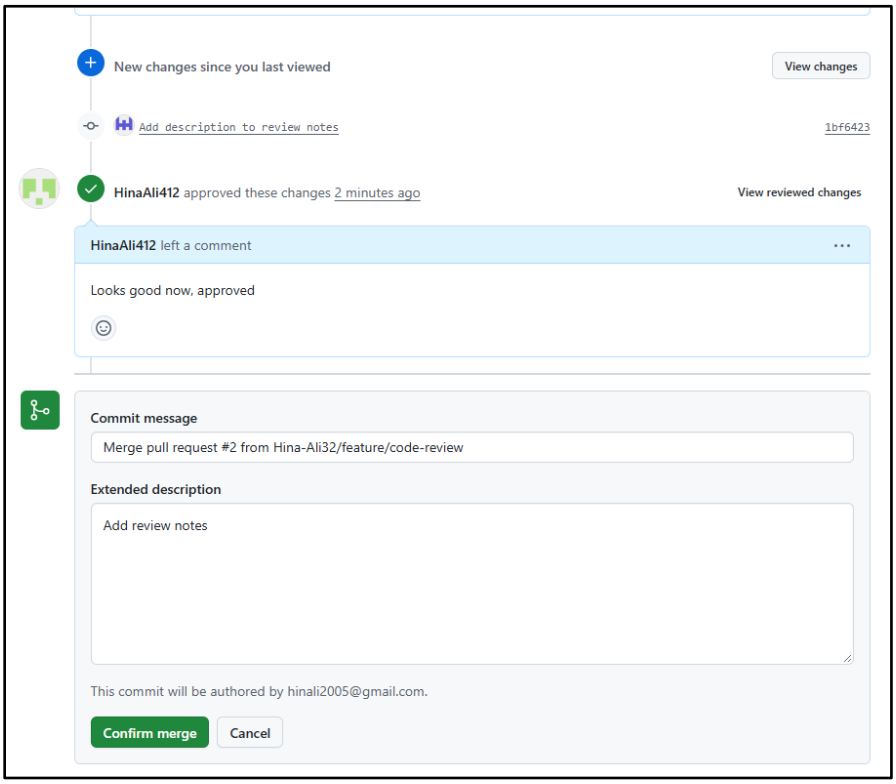
Shows the practice pull request #3 “Practice PR to be rejected” closed by the reviewer with a comment, marked with the red **Closed** badge.



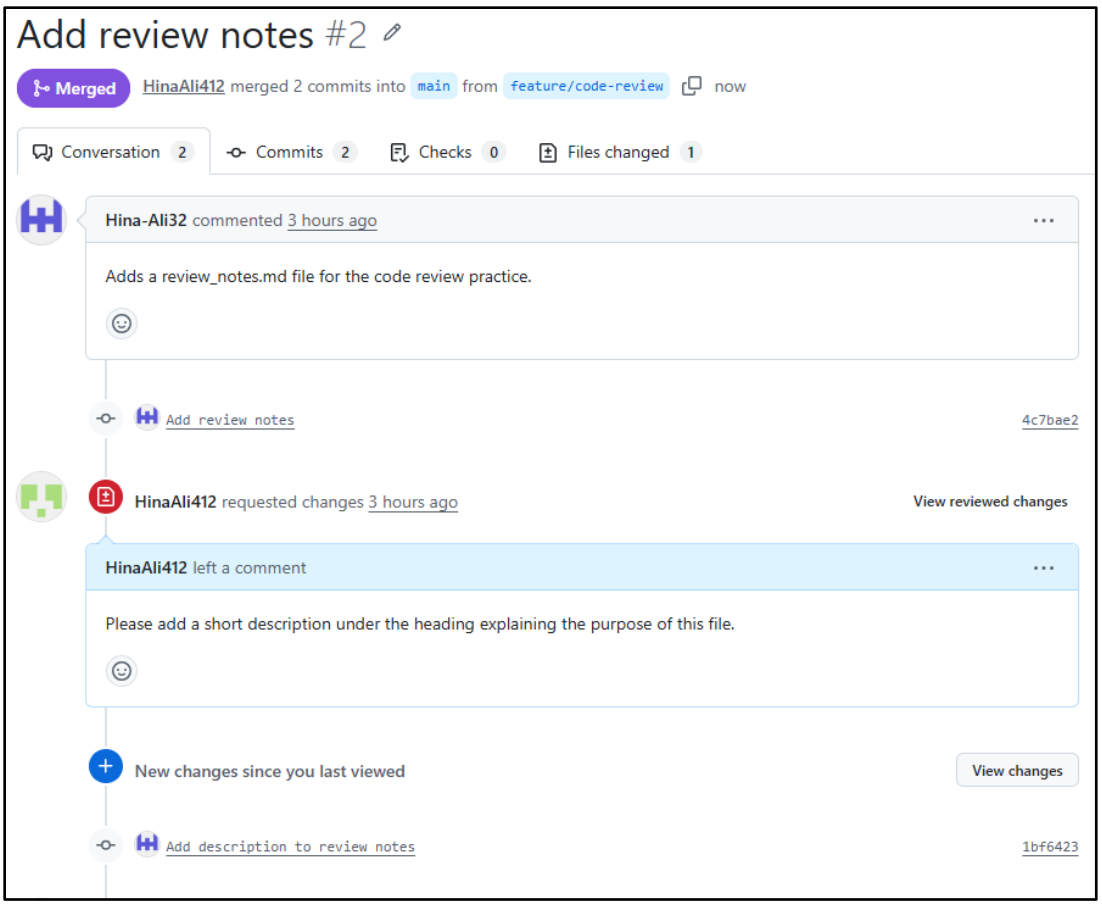
Shows pull request #2 updated automatically with a second commit, “Add description to review notes”, after the fix was pushed to the same branch.



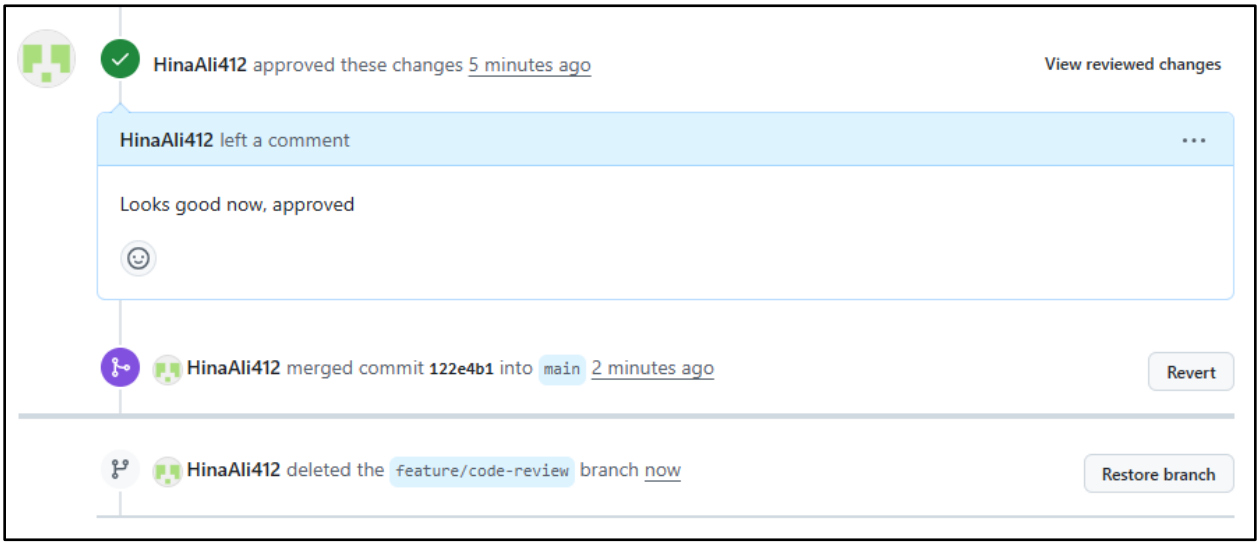
Shows the reviewer's approval and the **Confirm merge** box with a merge commit selected as the merge style.



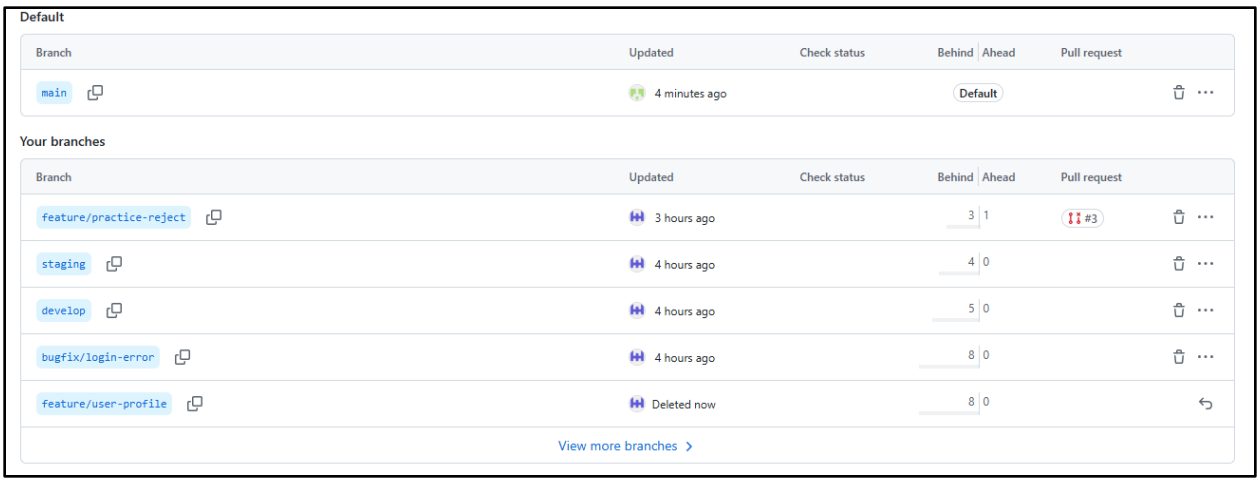
Shows pull request #2 merged into main by the reviewer, with the purple **Merged** badge.



Shows the feature/code-review branch deleted after the merge, with the **Restore branch** button visible.



Task 13: Branch Cleanup Best Practices

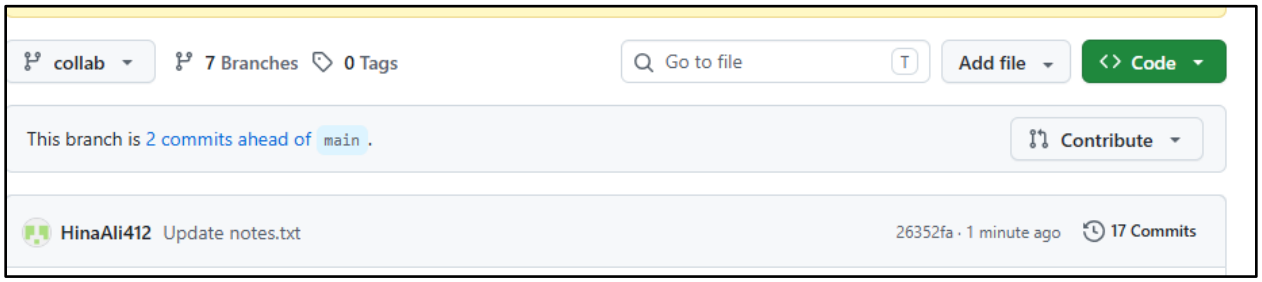


Shows the merged branch feature/user-profile deleted through the GitHub **Branches** page, marked "Deleted".

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin --delete bugfix/login-error
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
 - [deleted]         bugfix/login-error
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git push origin --delete <branch-name> removing a finished branch from GitHub, confirmed by the [deleted] line.

Bonus Task: Simulated Team Collaboration



Shows two commits on the collab branch from two different accounts, each adding a name and a fun fact to notes.txt.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git pull origin collab
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1.01 KiB | 1.01 MiB/s, done.
From github.com:Hina-Ali32/DevOps_Lab2_Practice
 * branch            collab     -> FETCH_HEAD
   5fbd43d..26352fa  collab     -> origin/collab
Updating 122e4b1..26352fa
Fast-forward
 notes.txt | 2 ++
 1 file changed, 2 insertions(+)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git merge collab
Already up to date.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   122e4b1..26352fa  main -> main
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

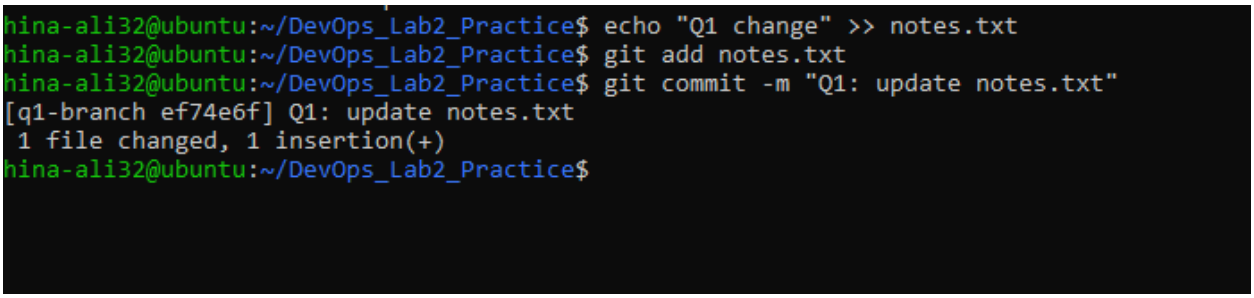
Shows collab merged into main and pushed with git push origin main.

Q1: Branching and merge verification

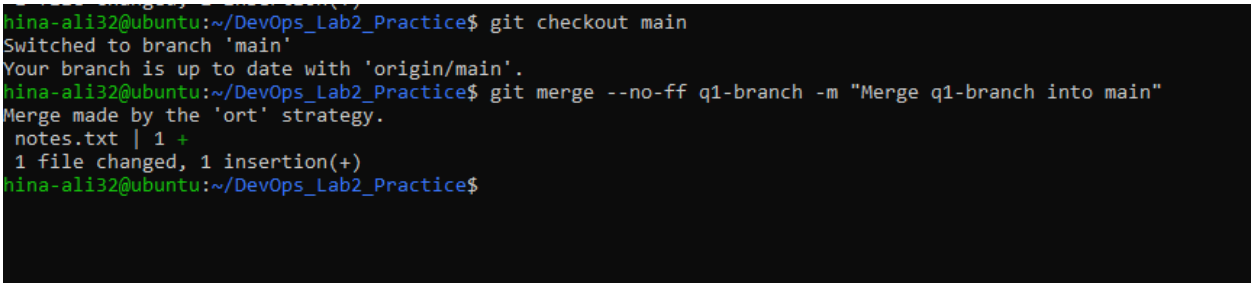
Shows the new branch q1-branch created with git checkout -b.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ cd ~/DevOps_Lab2_Practice
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git pull origin main
From github.com:Hina-Ali32/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
Already up to date.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b q1-branch
Switched to a new branch 'q1-branch'
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows a small change to notes.txt committed on q1-branch with the message “Q1: update notes.txt”.



Shows q1-branch merged into main with git merge --no-ff, creating a merge commit.



hows git log --oneline --graph with the commit “Merge q1-branch into main”, where the branch lines split and join, verifying the merge.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline --graph -5
*   e6bf929 (HEAD -> main) Merge q1-branch into main
|\
| * ef74e6f (q1-branch) Q1: update notes.txt
|/
* 26352fa (origin/main, origin/collab) Update notes.txt
* 5fbd43d (collab) Add main account name and fun fact
*   122e4b1 Merge pull request #2 from Hina-Ali32/feature/code-review
|\
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Q2: Multi-stage workflow

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Already on 'main'
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git branch
  bugfix/login-error
  collab
  develop
  feature-1
  feature/code-review
  feature/db-connection
  feature/practice-reject
  feature/user-profile
* main
  q1-branch
  staging
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows the three branches main, develop and staging listed with git branch.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout develop
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git pull origin develop
From github.com:Hina-Ali32/DevOps_Lab2_Practice
 * branch            develop    -> FETCH_HEAD
Already up to date.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b feature/q2-search
Switched to a new branch 'feature/q2-search'
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ echo "# Q2 search feature" >> q2_feature.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add q2_feature.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Add Q2 search feature"
[feature/q2-search d8dc8e3] Add Q2 search feature
 1 file changed, 1 insertion(+)
 create mode 100644 q2_feature.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows feature/q2-search created from develop with one commit adding q2_feature.md.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout develop
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git merge --no-ff feature/q2-search -m "Merge feature/q2-search into develop"
Merge made by the 'ort' strategy.
 q2_feature.md | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 q2_feature.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin develop
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 400 bytes | 200.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   bac02ec..97352e4  develop -> develop
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows feature/q2-search merged into develop and pushed.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git pull origin staging
From github.com:Hina-Ali32/DevOps_Lab2_Practice
 * branch            staging    -> FETCH_HEAD
Already up to date.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git merge --no-ff develop -m "Merge develop into staging"
Merge made by the 'ort' strategy.
 q2_feature.md | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 q2_feature.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin staging
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 224 bytes | 224.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   5bd3f37..dd265e2  staging -> staging
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows develop merged into staging and pushed

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git pull origin main
From github.com:Hina-Ali32/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
Already up to date.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git merge --no-ff staging -m "Merge staging into main"
Merge made by the 'ort' strategy.
 q2_feature.md | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 q2_feature.md
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 643 bytes | 321.00 KiB/s, done.
Total 6 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 2 local objects.
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   26352fa..d3a76bf  main -> main
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows staging merged into main and pushed, completing the workflow.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git branch --contains feature/q2-search
  develop
  feature/q2-search
* staging
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git show develop:q2_feature.md
# Q2 search feature
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git show staging:q2_feature.md
# Q2 search feature
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git show main:q2_feature.md
fatal: path 'q2_feature.md' exists on disk, but not in 'main'
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

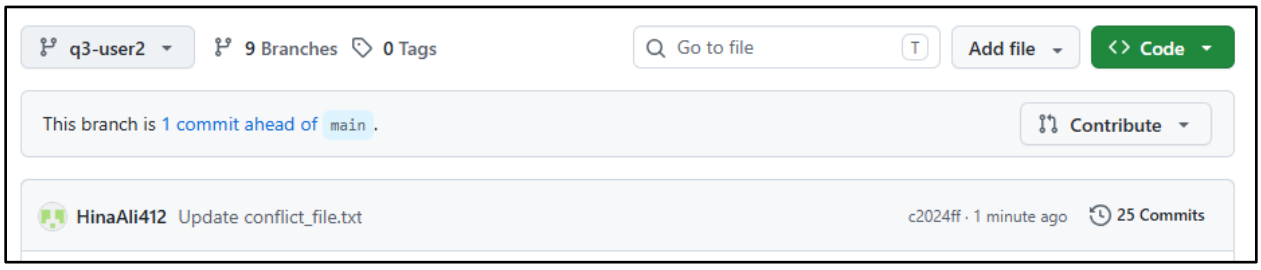
Shows git branch --contains feature/q2-search listing develop and staging but not main, git show printing the change from develop and staging, and an error for main, which proves the change passed through both stages before reaching main.

Q3: Collaboration and conflict resolution

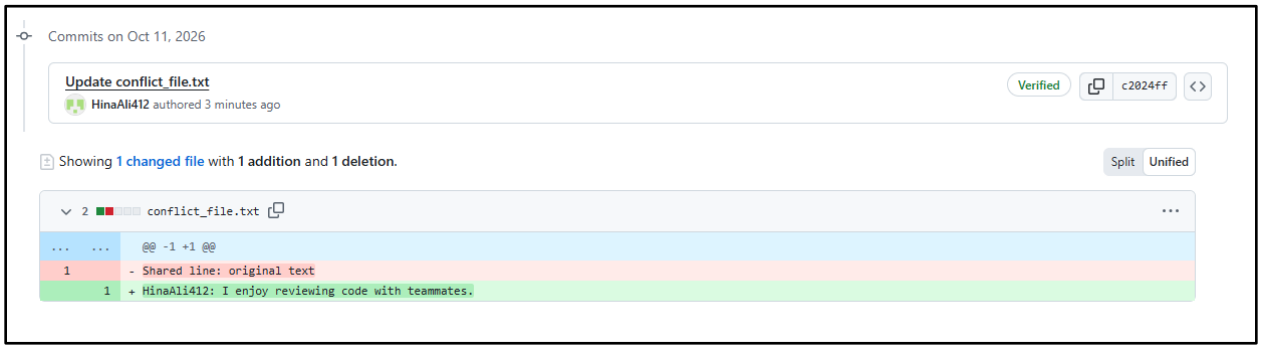
```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ cd ~/DevOps_Lab2_Practice
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git pull origin main
From github.com:Hina-Ali32/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
Already up to date.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ echo "Shared line: original text" > conflict_file.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add conflict_file.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Add conflict_file.txt for Q3"
[main 91d4625] Add conflict_file.txt for Q3
 1 file changed, 1 insertion(+)
 create mode 100644 conflict_file.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 317 bytes | 317.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   d3a76bf..91d4625  main -> main
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b q3-user1
Switched to a new branch 'q3-user1'
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ echo "Hina Ali: I prefer small, frequent commits." > conflict_file.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add conflict_file.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Q3: user1 changes the shared line"
[q3-user1 ee59b6a] Q3: user1 changes the shared line
 1 file changed, 1 insertion(+), 1 deletion(-)
```

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b q3-user1
Switched to a new branch 'q3-user1'
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ echo "Hina Ali: I prefer small, frequent commits." > conflict_file.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add conflict_file.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Q3: user1 changes the shared line"
[q3-user1 ee59b6a] Q3: user1 changes the shared line
 1 file changed, 1 insertion(+), 1 deletion(-)
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push -u origin q3-user1
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 318 bytes | 159.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'q3-user1' on GitHub by visiting:
remote:      https://github.com/Hina-Ali32/DevOps_Lab2_Practice/pull/new/q3-user1
remote:
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
 * [new branch]      q3-user1 -> q3-user1
branch 'q3-user1' set up to track 'origin/q3-user1'.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows conflict_file.txt created on main, then changed and pushed on the branch q3-user1



Shows the second account's commit on q3-user2, replacing the same line of conflict_file.txt with its own text.



Shows both branches, q3-user1 and q3-user2, in the repository's branch list.

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git fetch origin
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 952 bytes | 952.00 KiB/s, done.
From github.com:Hina-Ali32/DevOps_Lab2_Practice
 * [new branch]      q3-user2   -> origin/q3-user2
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git checkout q3-user1
Already on 'q3-user1'
Your branch is up to date with 'origin/q3-user1'.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git merge origin/q3-user2
Auto-merging conflict_file.txt
CONFLICT (content): Merge conflict in conflict_file.txt
Automatic merge failed; fix conflicts and then commit the result.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch q3-user1
Your branch is up to date with 'origin/q3-user1'.

You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
        both modified:   conflict_file.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        error

no changes added to commit (use "git add" and/or "git commit -a")
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ cat conflict_file.txt
<<<<<<< HEAD
Hina Ali: I prefer small, frequent commits.
=======
HinaAli412: I enjoy reviewing code with teammates.
>>>>>>> origin/q3-user2
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows git merge origin/q3-user2 failing with CONFLICT (content), and conflict_file.txt containing the conflict markers around both versions.

```
>>>>>>> origin/q3-user2
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ sed -i '/^<<<<<<< /d;/^=======$/d;/^>>>>>>> /d' conflict_file.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ cat conflict_file.txt
Hina Ali: I prefer small, frequent commits.
HinaAli412: I enjoy reviewing code with teammates.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git add conflict_file.txt
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Resolve merge conflict: keep both contributions"
[q3-user1 5b47e36] Resolve merge conflict: keep both contributions
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows the conflict markers removed, both lines kept, and the resolution committed as “Resolve merge conflict: keep both contributions”

```
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ cat conflict_file.txt
Hina Ali: I prefer small, frequent commits.
HinaAli412: I enjoy reviewing code with teammates.
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline --graph -5
*   5b47e36 (HEAD -> q3-user1) Resolve merge conflict: keep both contributions
|\
| * c2024ff (origin/q3-user2) Update conflict_file.txt
* | ee59b6a (origin/q3-user1) Q3: user1 changes the shared line
|/
* 91d4625 (origin/main, origin/HEAD, main) Add conflict_file.txt for Q3
*   d3a76bf Merge staging into main
|\
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$ git push origin q3-user1
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 392 bytes | 196.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:Hina-Ali32/DevOps_Lab2_Practice.git
   ee59b6a..5b47e36  q3-user1 -> q3-user1
hina-ali32@ubuntu:~/DevOps_Lab2_Practice$
```

Shows the final conflict_file.txt containing both collaborators’ lines, with the merge history graph and the push of q3-user1.